ZAKIR HUSAIN COLLEGE OF ENGINEERING AND TECHNOLOGY,
ALIGARH MUSLIM UNIVERSITY

Reinforcement Learning & Game Theory (AIE4230)

HOME ASSIGNMENT

NAME: SYED FAWAZ AYAZI

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FACULTY NO: 22AIB116

ENROLMENT NO: GN2417

SUBMITTED TO: Ms. Aysha Khan

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ASSIGNMENT

Q1 Stackelberg model of oligopoly - The Stackelberg model of oligopoly is a leader-follower model of oligopoly where one firm (the leader) chooses its output first, and the other firm (the follower) chooses its output after observing the leader's decision.

In this sequential setup, the leader gains a first mover advantage because it ~~on~~ can anticipate the follower's reaction and choose an output that maximize its own profit.

Follower's best-Response - Follower ~~behavior~~ behaves like a cournot competitor

$$\left[BR_F(q_E) = \frac{A - C - Bq_L}{2B}\right]$$

Stackelberg Equilibrium

$$q_L^* = \frac{A-C}{2B} \qquad q_F^* = \frac{A-C}{4B}$$

~~Eg Airbus (leader) + Boeing (Follower)~~

~~If airbus~~

leader earns more, higher profit follower produces less earns lower profit.

Eg Intel (leader) AMD (follower) in microprocessors

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Q2 Players → BPCL & HPCL

strategies → Each firm chooses quantity $q_F \geq 0$ (BPCL) or $q_N \geq 0$ (HPCL)

Total quantity → $Q = q_F + q_N$

Demand → $P(Q) = A - BQ$

c → cost per unit for each firm

Profit

$$\pi_F = q_F (A - B(q_F + q_N) - c)$$

$$\pi_N = q_N (A - B(q_F + q_N) - c)$$

Best response function

$$\frac{\partial \pi_F}{\partial q_F} = 0 \quad \& \quad \frac{\partial \pi_N}{\partial q_N} = 0$$

$$A - 2Bq_F - Bq_N - c = 0 \Rightarrow q_F = \frac{A - c - Bq_N}{2B}$$

$$A - 2Bq_N - Bq_F - c = 0 \Rightarrow q_N = \frac{A - c - Bq_F}{2B}$$

Cournot - Nash (symmetric) equilibrium

$$q_F = q_N = q^*$$

$$q^* = \frac{A - c - Bq^*}{2B} \qquad \frac{3}{2} q^* = \frac{A - c}{2B}$$

$$q_F^* = q_N^* = \frac{A - c}{3B} \qquad Q^* = q_F^* + q_N^* = \frac{2(A - c)}{3B}$$

$$P^* = A - BQ^* = \frac{A + 2c}{3}$$

For each firm

$$\pi_F^* = \pi_N^* = (P^* - c) q^* = (A - c)^2 / 9B$$

If $c = 0$ $\quad q^* = A/3B \quad Q^* = 2A/3B \quad P^* = A/3$

$$\pi^* = A^2/9B$$

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Q3 The Median Voter theorem says that in a majority voting system, competing parties choose policies preferred by the median voter.
When ~~voting~~ vote buying is introduced, the model changes: instead of appealing only to the median voter, a party can shift resources (money, subsidies, transfers) to certain voters to secure majority support.

A modified version works like this
→ Parties still choose a policy platform.
→ But they can also allocate targeted transfers to voters whose votes are cheaper to buy than shifting policies towards the median.
→ Instead of converging to the median voter's pref, Parties optimize the combination of policy + Purchased votes that give them 50% + 1 voters at min cost.

Thus with vote buying, policy must not match the median voter's ideal point - The outcome reflects stratigic buying of pivotal voters, rather than pure preference aggregation.

In the modified model, a party max votes using both policy choice (x) and transfers ~~t~~ to voters

A party wins if

$$\sum_{i \in N} (x, t) \geq \frac{N}{2} + 1$$

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where $V(x, t_i)$ is the set of voters who support the party given policy x and transfers t_i

A voter i supports the policy if

$$V_i(x) + t_i \geq U_i(x^{opp})$$

without vote buying

$$x = x_m \text{ (median voter's ideal point)}$$

with vote-buying

$$\min \sum_i t_i \text{ s.t. } \sum_i V_i(x) + t_i \geq U_i(x^{opp}) \quad 1 \geq N/2 + 1$$

This shows the party can move away from the median voter outcome and instead buy the cheapest combination of voters needed to form a majority.